

Parkinson's Disease Drug Repurposing — Screening Table (Final, Complete)

Status: Screening complete. 26/26 unique candidates screened (27 raw outputs, with Mitoxantrone merged across its two discovery branches per Step 1 identity-resolution rule). Per instruction, this document STOPS at screening — no Deep Validation has been performed.

Data source policy (frozen, applied to every group without exception): Monarch Initiative KG, PubMed (official web UI / E-utilities API only), ClinicalTrials.gov (official API only), Google Patents (sole primary patent-content source; Justia Patents for patent-number discovery only, never sufficient alone for an N2/P3 call). General WebSearch, PubChem patent pages, Espacenet, WIPO PATENTSCOPE, USPTO search, Wikipedia, vendor/blog/aggregator pages, and unsupported model knowledge were never used. A technical failure on an approved source is recorded as **SOURCE_ACCESS_FAILED/UNRESOLVED** for that specific claim — never silently compensated for by switching to an unapproved source.

Note on this consolidation: the 6 groups completed earlier this session (DOPAL, CMA, MCU/Ferroptosis, Sigma-1/MAM, Axonal transport/Golgi, Synaptic vesicle/Complement) are reproduced here **verbatim** from their original screening output — not rerun, not reinterpreted. Only the 7th group (Senescence/SASP) was newly run in this consolidation pass, after two prior dispatch attempts failed at the Agent tool's own safety-classifier gate (**claude-sonnet-5[1m] is temporarily unavailable (timed out)**) before ever reaching the research stage — a tool-layer issue unrelated to data-source access. The third attempt succeeded and its result is included below in full, also verbatim.

Master Screening Table (26 candidates)

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
Pyridoxamine	Pyridoxamine ("PM"); CHEBI:16410; CAS 85-87-0	DO PA L/dopamine -quinone oxidative toxicity	Direct nucleophilic scavenger of dopamine -quinone	Yes. PMID 41453945 (OBSERVED_LITERATURE, title/j	PMD 41453945 (OBSERVED_LITERATURE, title/j	N1	P2	—	—	—	—	—	—	—	REJECT	Exact compound already directly tested in a PD-associated	Full methods/abstract of PMID 41453942 unverified

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	(OBSERVE_D_K_G). Clinical name "Pyridoxine" returned no Pub		e (DAQ) and reactive carbonyls	parameter-induced alphasynuclein oligomerization through sc	our annual/ye ar; full abstract S O U R C E _ A C C E S S _ F A I L E D)											ted cellular/molecular model (DAQ scavenging + α -synuclein oli	— access blocked at abstract-page level.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	bMed hits (auto-re directed to "pyridoxine"). Name-collision flag:			avenging neurotoxic dopamine quinone" (Sci Rep 2025)												gomerization) → N1/P2 stop rule.	

[illegible]

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	a different vitamer.																
L-carnosine	Betalanin-L-histidine; CHEBI: 15727; CA	DOPAL/doamine-quinone oxidative tox	Direct carbonyl aldehyde scavenger (Mi	Yes, extensively. PMID 33052927 (6-	4 PMIDs (OBSERVED_LITERATURE)	N1	P2	—	—	—	—	—	—	—	REJECT	Exact compound already directly and re	No significant — well-supported by mu

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	S305-84-0; Drug Bank DB11695 (OBSERVED_KG).	icity	chael addition) + secondary Nrf2 engagement	OHDA in vitro PD model); PMID 34138535 (Thy1- α Syntra												peatedly tested across multiple PD animal/cell models → unambiguous	Multiple independent hits.

[illegible]

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				D37371945 (rotation rate PD model).													
Bismuth subnitrate	Colloidal bismuth salts.	DOPAL/dopamine-q	Metallotionein (MT1/MT2)	None found. No PubMed	PubMed "bismuth AND"	N3	P4	M2	F1 (AGENT_INFERENCE)	—	—	MT induction in astrocytes	If real, plausibly inters	Documented human	HOLD	Passes novelty (N3/	No study has tested the

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ate / substrate	Not separately indexed in Monarch; nearest relative is bismu	union of oxidative toxicity	T2) induction via MTF-1/MRE, hypothesis sized to protect DANE	ed record combinationing any bismu compound with a direct PD ex	Parkinson " (10 hits, all toxicity/encephalopathy, no ne exact				NCE bridge — no exact-compound functional phenotypic demo			ytes by generic bismu therapeutic exposure (AGENT_INFERENCE	ects the independent y-supported MTPDneurorotect ion axi	bismuth-salt neurotoxicity: myoclonic encephalopathy at ex ce		P4); mechanistic target-clas supp ort real and rep ate d	exact salts specifically for CNS MT induction /DANEurorot

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	th sub ali cyl ate (CHEBI: 261649). Name-collision flags: (1) bis mu		ur on s from DAQ toxicity	pe rim ent al or the ra pe uti c tes t.	-comp ound PD experiments); PMID 8891274 (bismuth induces				nst rat ed)			sal t ide ntit y un co nfir me d)	s (AGEN T_INFERENCE)	ssi ve ex po sur e (PMID 2566254, 3484854)		(M2), but exact-comp ound functional evaluation evidence only F1/inf erent	ect ion ; Clin ical Tria ls.g ov an d Go ogl e Pa ten ts un re ac ha ble ,

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	th-210 radioisotope hits unrelated; (2) most "bi smuth +Park ins				MT in rat astrocytes, not PD-specific) ; target-classes MT-PDlite											ial, plus a documente d human CNS-toxicity liability. Does not me	so a P3 /patent rejection cannot be fully ruled out .

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	on "hit" concern bis-muth-inducedencephalopathy/myoclon				rat ure (P MID 17 88 29 9, 12 67 13 04, 34 62 32 11, 40 75 40 45);											et F2 /F 3 preferred-profile bar.	

[illegible]

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
C A7 7.1	Successor to AR 7/Q X3 9/Q X7 7; RA Rα -correspondent interaction	Chapter on e-mediated autophagy (LAM P2A-dependent) de	RARα-corepressor complex → transcriptional repression	None found (no PubMed record of any kind).	Monarch: RARα ↔ NCR1/NCOR2 (OBSERVED_KG); C	N3	P4	M2	F0	consistent (AGENT_INFERENCE, target-class only)	uncertain	None found for exact compound	N/A	None found	HOLD (low priority)	Target pathway is genuinely PD-relevant, but the exact compound	Whether CA 77.1 has ANY published data (even non-PD)

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	ion disruptor. Zero hits in PubMed, Monarch, and Justia for	cline in aging/ PD	of LAMP2A		MA/LAMP2A-PD target-class literature (PMID 29949106,											has zero retrievable footprint anywhere — F0, identity unco	demonstrating LAMP2A/CMA activation is entirely unverifie

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	"C A7 7.1 " and predicted essor codes. Identify itself unverified				28066181, 27173998); "C A7 7.1 "/" AR 7"/ "Q X3 9"/ "Q X7 7" = 0 hit											nfir mable . Treat ed as an evi de nc e gap, not a dis co nfirm ing fin	d; ori gin al AR 7/ Q X3 9/ Q X7 7 pu bli cat ion not loc ata ble via ap pr

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	able through the four approved sources — high risk of				severely where											ding.	oved sources.

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	an unpublished/ internal compound code.																
Dihydroomycine	CHEBI: 28917; syn	Chaperone-mediated	LAMP2A/HS C70	Yes — extensive	6 PMIDs, all O	N1	P2	—	—	—	—	Neuroprotective	Direct — the same	None found (one un	REJECT	Exact compound alr	N/A (rejected on no

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
tin (DH M)	n. ampelopsin; flavanone, 2,3-dihydro derivative of myricetin	ted autophagy decline in aging/PD	CMACHINE — natural flavonoid CMA enhancer	. MPTP mouse model (PMD 27374489); MPTP chronic	BSERVED_LITERATURE, exact compound							os multiple direct PD models	PD models	related retracted DHM paper not part of the PD evidence		easy directly tested in multiple PD animal/cell models including a	velty) — absence of human PD trial data if ever revisited.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	n. No collision risk.			mouse, pyroptosis/necroptosis (PMID 37712262); T2DM rat										base)		study on the exact proposed mechanism. Not a novel repurposing	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				PD-like model (P MID 36859835); sal solino l-induced DA toxicity												ng hypothesis.	

[illegible]

[illegible]

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
				on and enhance CMA" (PMID 31223479).													
Saivianoli	CHEBI: 13	Chaperon	LAMP2A/	Yes — ex	5 PMIDs	N1	P2	—	—	—	—	Neuroprot	Direct — the	PMID 20	REJECT	Exact comp	N/A (rejec

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candidate (SAB)	4301; CAS 121521-90-2; syn. Dan Shen Suan B / Lithosperm	e-mediated autophagy decline in aging/PD	HS C70CMACHINE — natural ur al polysaccharide	tensive . 6-OHDA-induce d apoptosis (PMID 174684	OVERD_LITERATURE (1 additional P MID 20686228C							ective/antioxidant across multiple direct PD toxins models	se ARE PD models	686228retacted (flagged, excluded); independent findings unaffected		ound already directly tested in multiple PD cell/toxin models over	ted on novelty) — absence of human PD trial/exposure data

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	mic acid B; polyphe nol from <i>Salvia miltiorrhiza</i> . No collision		nc er	); MPP +-induced injury (PMD 33 99 45 27); Nrf2-dependent	ONTRA DICTED — retracted, not relied upon)									ected		a decade, plus the same paper demonstrate the exact propos	despite ~15 years of preclinical work.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component und functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	risk.			glial protection (PMID 24991814); A53T α -synuclein amyloid												ed CMA/LAMP2A mechanism jointly with DDM. Not no	

[illegible]

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				dy with DHM (PMID 31223479).													
Vitamin K1 (Phyllo	CEBI: 18067; CAS	Ferroptosis/lipid perox	FS P1 (AlF M2) redu	Pu bMed "phyloquinon	P MID 23064431; P	N1 (tentative)	P2	—	—	—	—	STOP at Step 2	—	STOPPED — N1/P2	REJECT	Pu bMed independentl	Abstract text for both

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quinone)	84-80-0; syn. phyto menadione (OBSERVE_D_KG). Collision	idation, FS P1-C oQ 10 axis	cess vitamin K → hydroquinone radical-rapping antioxidant	e AND Parkins on " = 2 records: PMID 23064431 (vitamins K	MI D 22582012 (title/s earch-match h OBSERVE_D_LITERATURE											y surfaces two records testing vitamin K species directly in	PMDs could not be retrieved — need to confirm phyloquinon

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	k HLG — literature freely mixes K1/K2/K3, pharmacol		(general mechanism, non-PD, PMD 3592516)	interact with N-terminal α -synuclein, modulate fibrillization in	; full abstracts S O U R C E _ A C C E S S _ F A I L E D)											PD-as sociated molecular/genetic models, triggering the determi	especifically (not only K2/K3) was one of the tested sp

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	logically distinct.			vitro) and PMID 22582012 (vitamin K2 rescues pin k1 deficiency												nistic stop rule.	ecies.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				cy) — both PD-associated molecular/r/genetic models.													
Oltipraz	CH EBI:	Ferropto	NQ O1	None found	PubMed	N3/N4	P4	M3	F2	Consistent	Feasible	Neurop	High —	None found	KEP (Pr	No exact	CT.gov

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	77319; CAS 64224-21-1; syn. RP-35,972 (OBSERVE_D_KG)	sis /FSP 1-CoQ10 axis, parallel NQO1 route	via Nrf2/Keap1 induction — "a novel activator of Nrf2" (PubMed)	nd. All PubMed/Justicia queries = 0 results (repeated).	0 hits (multiple queries); Justia 0 hits; CT.gov SOURCE					stent		rotective (reduced oxidative stress/oxidative stresses/ion overload/ferritin	same oxidative-stress/ferritinosis in PD; NQO1-PD link	nd; non-PD clinical tolerability history not verifiable (C.T. global	ef err ed)	-compound PD evidence or patient anywhere (true novel-pool	inaccessible — cannot confirm/rule out multiple trials or surfac

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
					_ACCESS_FAILURE							cerebral-ischemia/stroke animal models via Nrf2/NQO1 activation	kind independently validated (MPTP model P MID 33271195; generated	ed)		indicate); NQO1 independently and repeatedly linked to PD	clinical safety signals.

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component of functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC/HOLD	Reason	Most important remaining inference gap
												ivation	netic meta-analysis PMID 25054016; Akt-NQO1 PD mechanism			including in PD animal models; CNS-active, ferroptosis-protecti	

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
													P MID 31 35 86 53; Nrf 2-Keap1 PD model P MID 38 45 50 26)			ve phenotype outside PD .	

Ca nd id at e	Ca no ni cal id en tit y	Di sc ov er y br an ch	Ta rg et/ ac tio n	Ex ac t pri or PD evi de nc e	Ev id en ce so ur ce	N	P	M	F	Dir ec tio n	C NS	Ex ac t-c o m po und fun c tio na l ph en ot yp e	PD rel ev an ce of ph en ot yp e	Co ntr ad ict or y evi de nc e	KE EP /R EJ EC T/ H OL D	Re as on	M os t im po rta nt re mai ning inf er en ce ga p
Mi to xa ntr on e (m er ge d — tw o br an ch es)	C HE BI: 50 72 9/5 07 27; CA S 65 27 1- 80 -9; Dr ug Ba nk DB 01 20 4;	(a) M C U/ Ca ²⁺ - ov erl oad — ne gat ive mo dul ator ; (b) US P1 1 de ubi qui tin ati	(a) M C U co mp lex ne gat ive mo dul ato r; (b) US P1 1 de ubi qui tin	No ne fou nd for PD . Pu bM ed "mi tox ant ro ne Pa rki ns on " (5 hit s)/	(a) P MI D 34 16 17 74 (F DA -lib rar y M C U scr ee n), P MI D 38	N3 /N 4	P4 (P D); P0 (M S, se pa rat e ind ica tio n)	M2 (br an ch a: M3 ; br an ch b: M1 /M 2)	F0 /F 1 — pr es en ce in the cit ed fun cti on al as sa y un co nfir	(a) Co nsi ste nt; (b) Un cle ar	Fe asi ble -wi th- ca uti on (C NS -ac tiv e in M S via infl am ed/ dis ru pte	Un res olv ed — co uld not co nfir m mit ox ant ro ne was the tes ted co mp	N/ A pe ndi ng co nfir ma tio n	W ell- do cu me nte d do se- lim itin g car dio tox icit y (P MI D 15 62 36	H OL D	Re al tar get -cl as s rati on ale , no dir ect PD ex po sur e fou nd, but ex	(i) CT .go v ac ce ss to rul e out pri or PD tria ls; (ii) co nfir m mit ox ant

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	brand No vantage (OBSEVE D_K G). FDA-approved for M	ng). (b) Gologram entation — US P11 (opposes ITCH) axis.	ase	"dopamine rgic" (8 hits): all incidents, no PD-model tested. Justi	007529 (title only, MC U-ferritinosis); (b) PMID 41655130 (titl				med		d BBB; not proof of intact-BBB penetration)	ound in either cited functional paper		67) — systemic safety liability, real barrier for chronic PD do		act-com pound functional evidence unresolved for both branch	rone was the compound in PMID 38007529's functional as

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	S/certain leukemias (P0 for MS, not PD).	Both preserved.		a 0 patents.	e only; mitoxantro ne-U SP 11 link not independently confirmed									sing		es, branch (b) link looks inferred, CT.gov un checked, and car	say; (iii) confirm its role in PMD 41655130's Glogibranch.

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					d, AGENT_INFERENCE); CT.gov SOURCE_ACCESS											diotoxicity is a real chronic-dosing liability.	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
					_F AILED												
Ru 265	Cell-permeable Ru 360-derived MCU inhibitor;	MCU/Ca ²⁺ -overload	Cell-permeable ruthenium MCU pore blocker	No ne found. PubMed/Justia = 0 hits (repeated).	PubMed hits; Justia 0 hits; CT.gov S O U R	N3 /N 4	P4	M3 — MCUPD link independent ly strong (P	F2 — PMID 32126877: preserves cortical neuron	Consistent for core mechanism, wit h flagged ca	Feasible — directly demonstrate neuronal	Preserves cortical neuron mitochondrial respiration	High — mitochondrial bioenergetic protection /	P MID 38779706: documented off-target effects	KEP (Preferred)	No exact-compound PD evidence or patent found; str	Off-target excitability/neurotransmission profile

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	no CHEBI/Monarch entry; CASSUNRESOLVED. Collision risk		(PMID 34042919)		CE_ACESS_FAILED			MI D 27859782	on respiration after let halox ygen -glucose deprivation (exact co	ve at: PMID 38779706 — off-target increase in neuronal ex	and active in cortical neurons	on after OGD injury (F2)	Ca ²⁺ -overload prevented ion directly relevant to PD's core mitoc	on neuronal excitability/neurotransmission		on g independent target-clas s PD linkage (genetic + toxin PD	(PMID 38779706) need s characterization ; confirm wheth

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	LOW.							sh PD models; PMID 37086674 paraquat/PD neuro pathology)	compound, mitochondrial/bioenergetic phenotype, OGD/stroke	citability/reduced neuronal transmission			mitochondrial pathology			models); exact-compound CNS-neuroprotective phenotype co	er Ru 265 was the compound in PMID 38007529's ferropto

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									e model)							nfirmed out side PD ; CNS activity directly demonstrated.	sis as say.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
Benzethonium (chl oride)	Quaternary-ammonium topical antiseptic/preservative; not in Mo	MCU/Ca ²⁺ -overload	MCU negative modulator (FDA-approved library screen; not	None found. PubMed hits; Justia hits; Justia hits.	PubMed hits; Justia hits; CT.gov SOSOURCES ACCESS	N3/N4	P4	M3 — same independent M C U-PD target-class literature as	F0 — no exact-compound functional phenotyp e data	Unclear — no downstream specific confirmed concern	Uncertain, with a specific confirmed concern	None (F0, per rule)	N/A	PubMed search for systemic absorption/PK returned only top	HOLD (low priority)	Target-class rationales solid and novelty satisfied, but F0 plus co	No approved source established any systematic/CNS exposure

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
	narch K G; CAS/CI D U N RE S OL V E D. Collision risk LOW (di		independently named in retrievable title — SUPPORTED_C		_F A I L E D			Ru 26 5	ee n-hit identification					ical/dermal-delivery literature — no systemic or CNS bioav		confirmed absence of any systemic/CNS exposure literature	route for benzethonium — the single largest gap; dedicated PK

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	distinctive name).		CONTEXT, not directly confirmed)											ail ability data at all, consistent with real-world topical-only use		e makes this low-priority per the prioritization rule.	/BB data needed before reconsidering.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
PRE-084	Sigm a-1 receptor agonist reserch-to ol comp ound (2-(4-morp	M A M tethe ring (V AP B-PT PI P5 1) / ER -mitochondria Ca 2+	SIGMAR1 agonist, MAM-resident IP3R3 channel	Directly tested (0.3 mg/kg/day, 5 wk) in 6-OHDA intrastriatal-lesi	P MID 24755275 (OBSERVED_LITERATURE, abstract fetch	N1	P2	—	—	—	—	n/a (STOP at N1)	n/a	n/a	REJECT	Exact compound already directly and successfully tested in an	Whether PRE-084 has ever been given to human PD patient

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	holinoethyl)-1-phenylcyclohexanecarboxylate HCl) . No CA	-lipid transfer		on mouse model of parkinsonism ; for elimination-use recovery , inc	ed verbatim)											animal PD model with a mechanistically-linked functional read	ts (P1) is unresolved — CT.gov SOURCE _ACESS _FAL

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	S/ClD resolvable. Low collision risk.			read striatal DA fiber density, BDNF/GDNF upregulation; eff												out.	LED throughout; doesn't change the REJECT outcome.

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				ectabolished in Sigma-1-KO mice.													
Priodipidine	Also AC R16/Huntexil;	MAM tethe ring (V	SIGMAR1 agonist,	Directly tested in 3 individuals	3 PMIDs (titles OBS	N1	P2 (human P1 UNRE	—	—	—	—	n/a (STOP at N1)	n/a	n/a	REJECT	Multimodal (macaque,	Whether priodipidine

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	originally developed for Huntington's disease/chorea; extensive	APB-PTPIP51)	enhances MERCS number/IP3R-1R colocalization	ependent animal PD models: parkinsonism animal cases (reduce	ERVED_LITERATURE; full abstracts SOURCES_ACCESS		SOLVED — CT.gov inaccessible)									mouse, rat), multiple direct PD animal-model testing history	was ever described in a human PD trial (vs. only HD) remains

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	e HD clinical history (P MID 28 67 11 35, 29 12 02 64). No PD			s L-DOPA dyskinesia , P MID 30 57 59 96); mouse PD mo	_F ALLED on 2 of 3)											y already in the literature — unambiguous N1 .	ins UNRESOLVED — CT.gov inaccurate possible throughout .

[illegible]

[illegible]

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				or extra-terials).													
Cutaneous (S A4 50 3)	Signal-recognition; also the PET radio lig	MAM tetra ring (VAPB-PTPI P5 1)	SLG MAR 1 agonist, MAM/PI P3 R3-linked Ca2+	[11C] SAR 4503 PET directly administered to and	P MID 160085; P MID 32347983 (OBSERVE	N1	P1	—	—	—	—	n/a (STOP at N1)	n/a	n/a	REJECT	Exact compound already directly administered to	Full quantitative results and whether cutaneous

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	and [11C] SA 4503. No collision risk.		signaling	imaged in living human PD patients to map sigma-1 receptor de	D_LITERATURE, snippet-level; full abstracts SOURCES_A											human PD patients (as PET tracer) specifically to interrogate	was ever given the repeatedly (not only as PET microdose)

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				nsity and its relation to PD pathology (PMID 16008536), cor	CCESS_FAILED)											PD pathology — the strongest form of N1/P1 exclusion.	to PD patients could not be confirmed beyond search-snipet level.

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				83).													
Ne flamm ap im od (V X-74 5)	CAS/Cl UDUNRESSOLVED (not indexed in Monarch	Axonal transport/microtubule-motordysfunction	p38 α MAPK (MAPK14), ATP-competitive inhibitor (PMI	None found for PD. "neflamapimod AND Parkinsons on dis	PubMed Web UI; Justia (4 patents, formulation/co-crysta	N3	P4	M3	F3	consistent (AGENT_INFERENCE)	feasible (human CNS trials in DLB with plasma p-tau	Strong human translation al data (Phase 2B DLB trials	p38 MAPK strongly linked to PD dopaminergic pat	No ne found	KEEP	No direct compound-PD exposure of any kind; target	CT.gov access failed — could not independently confirm

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	). Low collision risk.		D 36 13 09 46)	ease" = 9 hits, all DL B-focused (Rewin D-LB Phase 2B trial, P	I only, no mention PD)						target-ing agreement response, PMID 37 65 79 39)	l; plasma aptau biomarker response), but entirely in DLB, not PD	hologymatching this exact branch (PMID 40 32 88 04 axonal			has strong independent PD mechanismistic support matching the	absence of a PD-labeled trial arm or PD-overlap in the DL

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
				MI D 38 70 62 71) mentioning PD only tangentially — DL B ≠									transport/AMPK-p38MAPK; PMID 26 26 17 96 review; mu			stated discovery branch; compound is CNS-penetrant/clinical	Btrial cohorts.

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				PD . Ne flama pi mo d-r as agi line hy brid (P MI D 38 41 24 54) is									Itip le 6- O H DA /ro ten on e pr ote cti on pa pe rs)			y ch ar act eri ze d (ju st not in PD).	

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
Tubastatin A	CAS 1252003-15-8; CHEBI: 94186 (OBSERVED_KG). No me	Axon al trans port/ micro tubule motor dy sfunction	HDAC6, selective inhibitor, ↑ α -tubulin-K40 acetylation	Yes — direct. Tested in vivo in multiple PD-associated model	4 PMIDs (search-nippet level; full abstracts SOURCES_A C	N1 (STOP)	P2	—	—	—	—	—	—	—	REJECT	Exact compound already directly and repeatedly tested in cell	Full abstract text of the four PMIDs inaccessible; recommend ma

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	an ingful collision risk.			s: 6-O H DA rat (P MI D 28 82 38 93, 37 56 07 27), M PP + ze br	CE SS _F AI LE D)											ar/ ani ma l PD models — fail s no vel ty out right at N1 .	nu al co nfirm ation of ex act do sin g/e nd poi nts if rev isit ed.

[illegible]

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				32296327).													
Dorsomorphin (Compound C)	CHEBI: 78510; CAS 866405-643; Drug Bank	Axonal transport/microtubule-motor dysfunction;	AMPK (PRKAA1/2), ATP-competitive inhibitor;	Yes — direct. PMID 32860006; Monarch CHEBI: 78510 (off	PMD 32860006; Monarch CHEBI: 78510 (off	N1 (STOP)	P2	—	—	—	—	—	—	—	REJECT	Exact compound already directly tested (as phar	Its use was as a mechanistic blocking tool,

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	nk DB 08 59 7. Name-e-collision risk H1GH — "Compound	nction	not selective — also a B M P/AL K2 (bone morphogenetic	morphin explicitly as the A M PK-inhibitor tool compound in rot	-target confirmation)											macol o g i c a l t o o l) i n a P D -a s s o c i a t e d c e l l u l a r m o d e l — f a i l s n o v e l	not a t h e r a p e u t i c t e s t , b u t t h e d e t e r m i n i s t i c N 1 r u l e m a k e s n o

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP /R EJ EC T/ H OLD	Reason	Most important remaining inference gap
	C" is a widely reused generic lab name, verified via Mona		prote in) signaling inhibitor (confirmed via Monarch CHEBI record +	enone-injured SH-SY5Y cells — a standard PD cellular mo												ty at N1 ; also carries a confirmed BMM P/AL K2 off-target select	such distinction; full abstract of PMID 32860006 not independent

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	rch /Pub Med to match do rso morphin unambiguously in the citati		P MID 31 36 38 85)	del .												ivit y liability.	ntl y confirmed.

Ca nd id at e	Ca no ni cal id en tit y	Di sc ov er y br an ch	Ta rg et/ ac tio n	Ex ac t pri or PD evi de nc e	Ev id en ce so ur ce	N	P	M	F	Dir ec tio n	C NS	Ex ac t-c o m po und fu nc tio na l ph en ot yp e	PD rel ev an ce of ph en ot yp e	Co ntr ad ict or y evi de nc e	KE EP /R EJ EC T/ H OL D	Re as on	M os t im po rta nt re m ai ni ng inf er en ce ga p
	on us ed.																
E mr ica sa n (ID N- 65 56)	CA S/ CI D U N RE S OL VE D (n ot ind ex ed in Mo na	Go lgi fra gm ent ati on str es s ph en oty pe	Ca sp as e- 2/3 , pa n-c as pa se inh ibit or (irr ev ers ibl e)	No ne fou nd. "e mri ca sa n AN D Pa rki ns on dis ea se" = 0	Pu bM ed we b UI; Ju sti a (8 pat ent s, for mu lati on/ de ute	N3	P4	M2	F2	co nsi ste nt (A G EN T_ IN FE RE N CE)	fea sib le (P MI D 29 10 31 02: sy ste mi c em ric as an re	Re du ce s ne ur on al inj ury /ap opt osi s in rat br ain isc	PD -re lev ant Go lgi fra gm ent ati on in DA ne ur on s ind ep	No ne fou nd dir ect ly co ntr adi cti ng; ext en siv e hu ma n	KE EP	No dir ect PD ex po sur e of an y kin d; pat hw ay co nn ect	Th e ca sp as e- 2/3 → G RA SP 65/ gia nti n cle av ag e

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	rch). Low collision risk.			results. General "emricasan" search (94 results) dominated	rated analog/flavivirus/somastostatin/macrophage/cell-culture						duced infarct volume/neurological deficit in rat brain ischemia-	hemiparesis/functional (PMID 29103102) and protects Zika-infe-	endently well-documented (PMID 33078843, 37855309, 3476	livelihood safety data, indication-irrelevant to PD		steps have never been shown together with DA-n euro n Golgi	

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-comboun d functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
				by liver disease (matc hes its extensive trial history), plus co	e, no mention PD)						ep erfusi on, implying functional al CNS exposure; no explicit BB-	cte d neur al cells (PMID 27 57 13 49) — both outside PD	95 17/ 23 93 93 44/ 22 17 77 21) — but that lite ratu re is Ra b1/ tra			ow s a relevant non-PD CNS prote ctive phenotype (F 2) with pla	fra gmentati on in the same experiment — the PD -relevance chain is

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				chlear cells, Zik a-infected neuronal cells, rat cerebral ischemia-r							PK data)		fficking-driven, not caspase-GRA SP65-driven; the caspase			usable CNS access.	assembled from two separate literatures; CT.gov inaccessible

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				ep erf usi on (P MI D 29 10 31 02) — no ne PD /do pa mi ne rgi c/s yn									→ G RA SP 65/ gia nti n me ch ani sm est abl ish ed onl y in ge ne ral (n				le to ver ify full hu ma n ex po sur e/ C NS -P K rec or d.

Ca nd id at e	Ca no ni cal id en tit y	Di sc ov er y br an ch	Ta rg et/ ac tio n	Ex ac t pri or PD evi de nc e	Ev id en ce so ur ce	N	P	M	F	Dir ec tio n	C NS	Ex ac t-c o m po un d fu nc tio na l ph en ot yp e	PD rel ev an ce of ph en ot yp e	Co ntr ad ict or y evi de nc e	KE EP /R EJ EC T/ H OL D	Re as on	M os t im po rta nt re m ai ni ng inf er en ce ga p
				ucl ein -sp eci fic.									on -P D) ap opt osi s lite rat ur e				
Bi s- T- 23	sy n. Ry ng o 1- 23; res ea rch too	Sy na pti c ve sic le en do cyt osi	D N M1 /2 GT Pa se, po siti ve all	No ne fou nd. Pu bM ed "Bi s- T- 23	Re sto res po do cyt e cyt os kel eta	N3	P4	M3	F2	co nsi ste nt	un cer tai n	Re sto res po do cyt e cyt os kel eta	Ph en oty pe is in kid ne y/p od oc	No ne fou nd	KE EP	N3 /P 4 (n o ex act -co mp ou nd	No ex act -co mp ou nd C NS /B BB

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component of functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	I compound, no CAS/CLD resolvable (UNRESOLVED). Name	sy dysfunction (dynamic/enveloping/fission)	osteric activation of self-assembly/GTPase activity	AND Parkinsons " = 0; Justicial/Google Patents = 0 relevant hits.	l/barrier function, ameliorates chronic kidney disease in cell+ a							l/barrier function in CKD models (non-euro-nal)	yte disease, not neuronal — relevance to PD by analogy (AGEN			PD exposure); M3 via independent monogenic PD-endocytosis	penetration data via any approved source; whether restoring dy

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	-collision risk: "bis-" is a generic chemistry prefix — verified				minimal models via dynamical oligomerization (PMLD 25 96 21)								T_INFERENCE for cross-tissue extrapolation)			netics (DNAJC6/auxilin, PMLD 33 98 36 93, 41 93 50 42, 26 52 89 54;	namination actually rescues the PD endocytosis defect (which

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
	target-specific hits only. Target confirmed: DN M1 (dynamic-1)				21, 27 43 27 39, 30 35 04 25, 41 07 72 41)											α-synuclein –A P2 /endo cytos interaction , P MI D 40 21 62 48,	h maps more to auxiliary-mediated uncoupling, a downstream step

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	activator.															25009269); F2 real activator phenotype in an independent no	p) is unverified.

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
																n-PD system; direction matches the branch's rescue logic.	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
Dynole 34-2	Research too complicated, no CAS/ClD resolvable (UNRES)	Synaptic vesicle endocytosis dysfunction	Dynamin I/II GTPase inhibitor	None found. PubMed "Dynole 34-2 AND Parkinsons" = 0; Justice	"Dynole 34-2, dynamin antagonist, suppresses endocytic F	N3	P4	M3	F2	inconsistent	uncertain	Direct, real confirmation of dynamin inhibition (not act	Direction contradicts the branch's rescue hypothesis	Yes — direction contradicts the branch's rescue hypothesis	REJECT	Same M3 pathway supported as Bis-T-23, but verified real-world evidence	None meaningful — the direction conflict is decisive; no plausible

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	OLVED). Target confirmed: dynamin I/II GT Pase-domain inhibitor.			a = 0.	M1-43 dye uptake" at the mouse neuromuscular junction —							ivation) at a real neuronal synapse		thesis. PMID 39786596 independently confirms suppression		nceshows the pharmacological direction is opposite to the pr	reformulated rescue rationalesurfaced.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
					blocks (not restor es) SV endocytosis in a live neuronal pr									n (not restoration) of SV endocytosis		opposed disease-rescue mechanism (blocks endocytosis rather	

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC/HOLD	Reason	Most important remaining inference gap
					ep ar ation (P MI D 39 78 65 96)											r than restoring it). A me ch ani sti c/t ool ref er en ce, not a via ble PD	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
																therapeutic lead.	
Pegcetacoplan	APL-2; FDA/EMA-approved 2021 as	Complete-mechanism-based synthetic proteasome inhibitor	C3/C3b, inhibitors C3 convertase / downstream	None found. PubMed "Pegcetacoplan AND	5 PMIDs on approved indications; CT.gov	N3	P4	M3	F3	consistent (in principle)	uncertain, learnability	Robust human clinical efficacy for its own	C3 blockade pharmacologically active and	No direct contradiction, but 0/0 hits on "peg	HOLD	Strong independent target-pathway evidence	Whether pegcetacoplan (or a reformulation)

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	Empaveli/Aspaveli (PNH, SC injection, PMID 38348608,	(C1q/C3-microglial)	eamopsonization-cl-eavage	Parkinsons = 0; CT.gov "pegcetacoplan Parkinsons"/neurod	vandJustin Aqueries returned genuine empty (not accepted							n approved (non-CNS) indications	dwell-tolerated in humans, but neither approved context is C	cetacoplan blood brain barrier/CNS" queries, and its only tw		dence for C3/complement in PD (PMID 34399786, 37704739,	lation, e.g. intrathecal) achieves any meaningful brain-paren

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP /R EJECT/ HOLD	Reason	Most important remaining inference gap
	39201383) and as Syfovre (intravitreal, geographic atrophy,			egeneration" = 0 studies; Justina = 0 patents (only a PNH patient	ss failures)								NS-relevant	o approved routes are SC (systemic, non-CNS target) and		41637879, 3928893) and strong human translation al F3	chyma exposure after systemic dosing — no PK /BB study

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	P MID 37 86 54 70, 31 47 44 39, 40 28 02 79). Peptide-classes C3			ent , US 20 23 00 76 52 7).										d intravital (local , into the eye — not systemic barrier-cro		data support KEP on M/F grounds, but the only validated CNS	located despite repeated retrie s.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	inhibitor (structure detail AGENT_INFERENCE). Neither ap													ssing)		-adjacent route is local intravitreal (never demonstrated to cro	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-component functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
	proved indication is PD/CNS.															ssBBB systematically) — a real, class-consistent liability, not just	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
																t absent data.	
ANX005	Tanrupurba rt (INN, per NCT07020819	Compound-meditation synaptics pruning (C	C1q, humanized neutralizing Ab blocking	None found for PD .CT.gov (official AP I) "A	CT.gov (7 real trials, none PD); PMID 29	N3	P4	M2	F3	consistent	uncertain (major open question for a full-le	Phase 1b open-label trial reporting safety	Demonstrates real human CNS-disease	None found	KEP	N3/P4 confirmed across PubMed, CT.gov,	Whether the HD trials' "clinical activity" refl

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combination functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	trial title); humanized anti-C1q neutralizing monoclonal IgG4 antibody	1q/C3-microglial)	classical pathway upstream of C3 cleavage	NX005" query returned 7 real trials — ALS (NCT04569435),	202623; PMID41787606 (HDP phase 1b trial)						neurodegeneration; BBB/CSF-permeability ratio study located)	/PK/PD/clinical activity of ANX005 in Huntington's disease patients	translation of signal (HD), supporting biological plausibility for C			and patients; M2 via C1q-classical-pathway PD-relevant literature	ectodermis; genuine CNS/C/SF target engagement by a full-length antibody

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	ibody (Annexon). No clinical profile confirmed: "humanized			warm autoimmune hemolytic anemia (NCT04691570), health								ients (P MID 41787606, Mov Disord 2026)	NS complement-pathway engagement, though HD ≠ PD and			(e.g., microglial cGAS-TIN2-GC1q axis in Mn-induced	ibody or a peripheral/immune-mediated effect — no CSF-PK or

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	Anti-C1q Antibody for Treatment of Autoimmune and Neur			hy volunteers (NCT03010046), Guilla in-Barré ×2 (NCT040351									CNS-access mechanism not established			ur od eg en er at ion, P MI D 42 53 19 37) plu s ge ne ral C3 -d ow	BB B-pe net rati on stu dy re trie va ble .

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	degenerative Diseases" (PMID 29202623).			35, NCT04701164, plus NCT07020819/Tanruprubart), Hunti												instream PD literature; F3 from real human HD trial activity	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
				ngt on's disease (NCT04514367) — none for PD ; Justia = 0												data; direction consistent; CNS uncertain but not disqualifying	

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
				patients.												en the HD clinical signal .	
PMX205	Cyclic hexapeptide C5aR1 (C	Complete ment-mediated synapt ic	C5aR1, brain-pentrant agonist	None found. PubMed "PMX205 AN	Suppressed inflammation by glial	N3	P4	M2 — no PD - or C5aR1-specific	F2 — strong, repeated, independent	consistent	feasible (uniquely well-s uppo	Consistent, repeated CNS-p en	Extrapolated PD-relevance from	None found	KEP	N3 /P4 confirmed; F2 strong re	No study has tested PMX205 (or

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	D88) antagonist, related to PMX53; no brand name, still	pruning (C1q/C3-microglial)	stsuppressing C5a-driven microglial/astrocyte activation	D Parkinsons on " = 0; CT.gov (official AP1) "PMX205" = 0 studies	responses in an Alzheimer's mouse model (PMID 39147742);			PD literature exists; pathway supports on the broader context	dent CNS functional evidence across multiple disease models		rtd among the five complete/dynamic candidate — co	etrant anti-euroinflammation completely efficacy across multiple independent	C5aR1's role in other neuroinflammation C5aR1's role in other neuroinflammation C5aR1's role in other neuroinflammation			peated independent C5aR1 functional evidence; direct consideration	C5aR1 blockade generally in an actual PD / α -synuclein model

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	preclinical (no CT.gov trials found at all) . No CAS/ClD res			s total; Justia = 0 patents.	neurorotation in hypoxic-ischemic brain injury (PMID 4196			implement-PD literature plus C5aR1's validated role in analogo	deletes		confirmed peripheral dosing reaching CNS targets, plus deduc	deficient CNS disease models (none is PD) — the strongest ex	from a direct C5aR1-PD data point			stent; CNS feasible — best-supported "P referred" profile in	— the PD-relevance inference is extrapolated from other neur

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	olved.				7875); attenuates neuroinflammation after intracerebral hemorrhage			using CNS neuroinflammation models			ated PK study)	act-compound functional CNS dataset of the branch				this branch.	oinflammatory CNS diseases.

[illegible]

[illegible]

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
					study (PMID 32064396)												
GSK-650394	Small-molecule SGK1 (serum/	Cellular senescence / SASP in	SGK1 in astrocytes/microglia	YES. Tested directly in microglia	PMID 33646633 (EMBO Mo	N1	P2	—	—	—	—	Direct, positive, multi-model	Direct	None found	REJECT	Exact compound already directly	GSK-650394 itself is a tool

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	glucocorticoid-regulated kinase 1) inhibitor; also reported as	PD glioma		ndopamine neuron/gliacultures (α -syn PF and SNCA	Med 2021; Author Correction PMID 40790103), OBSER											validated in the exact discovery-branched mechanism (glial S/GK1	compound with poor drug-like/PK/CNS-optimization properties

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP /R EJECT/ HOLD	Reason	Most important remaining inference gap
	an Aurora-B kinase inhibitor in unrelated oncology work (mi			-over expression models) and multiple in vivo PD animal mo	VED_LITERATURE											inhibition preventing senescence, protecting DA neurons) in cel	es; whether a clinical-grade, BB-B-optimized SGK1 inhibitor

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotype	PD relevance of phenotype	Contradictory evidence	KEP/R/EJ/EC/T/H/OLD	Reason	Most important remaining inference gap
	nor selectivity/collision flag, same compound). No CAS/CID			del; protected DA neurons, reduced pathological α -synuclein												lular and animal PD models — fails novelty at N1/P2.	could translate this already validated biology is the open question

[illegible]

[illegible]

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
				ch anism of this branch.													
Ruxolitinib	INCB 018424, FDA-approved JAK1	Senescence/ SASP — blocks IL-6/	JAK1/JAK2	No exact idiopathic c-PD study/t	Published "ruxolitinib Parkinsons" (4	N3	P4	M2	F2	Consistent	Feasible	Suppresses SASP/pro-inflammatory mediator	Direct mechanistic overlap: IL-6/JAK	None found; general caveat: JAK inh	KEP	Meets the full Preferred profile: P4 ,	No study has tested ruxolitinib direct

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combination functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	/JAK2 inhibitor (myelofibrosis, PV, GvHD, atopic dermatitis).	STAT3 SAP amplification loop		riall/modelfounded. One human case reported shows ruxolitinib	hits, none idiopathic-PD interventional); PMID 40501515 (CAR							ysecretome in secret cells, reverses frailty in aged mice (P	2/STAT3 axis drives gliosenescence and neuroinflammation is immunosuppressive (infection-risk consideration for chr			M2, F2, Direction = consistent, CNS =feasible, no direct PD ex	ly in a PD-specific model (MPTP/α-synuclein/genetic) to confirm

Candidate	Candidate entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	Monarch: CHEBI:66919. Also screened Round 2 under a diff			inhibitory effect for CA R-T- as associated parkinsonism in multiple	-T parkinsonism case); Monarch CHEBI:66919x MONDO:							MI D 26578790); reduce senescent neuroinflammation/improves out	mation, matching this branch's proposed PD pathology; multip	onically use), not a mechanistic contradiction		posur e — the strongest candidate in this branch.	reduced astrocyte/microglial senescence/ SASP markers in PD

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-comboun d functional phenotyp e	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/H OLD	Reason	Most important remaining inference gap
	erent (BM P6/7) framing: M2/unclear/feasible/hold — cited here			myeloma, but the paper itself distinguishes this iatrogenic mo	0005180 = 0 items; CT.gov SURCE _A CCESS _F AILE							comes in ischemic stroke (PMID 34367186), TBI (PMI	le CNS disease models show functional benefit from the same				-relevant glioma; whether standard approved or additional sing achieve

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	as SUPPORTED_CONTEXT only; classification below freshly			vement disorder from idiopathic PD — treated as SUPPORT	D; Google Patents SOURCES_ACESS_FALLED; Justice 0							D36813222), HIVencephalitis (PMID 26851503); effect	JAK-blockade mechanism				ves sufficient CNS exposure without unacceptable chronic im

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	derived for the SASP/IL-6/STAT3 mechanism.			ED_CO NT EX T, not counted toward N1/P1.	patients							ive in a human CNS parkinsons onism-like movement disorder					mu no suppression risk in elderly PD patients is untested.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
												(P MI D 40 50 15 15)					
Digoxin	CHEBI: 4551, cardiac glycoside from	Sevensence/SA SP — senolytic via alt	Na ⁺ /K ⁺ -ATPase	Not genuine intervention al drug -PD stu	P MI D 12856480, 12325312 (CON T	N4	P4	M2	F3	Unclear	Uncertain	Induces apoptotic senolysis via Na ⁺ /	Strong general senolytic mechanism	Direct mechanistic contradiction	HOLD	Genine senolytic mechanism with hu	Whether senolytic-ran gedigoxin ex

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	<i>Digitalis lanata</i> , FDA-approved Na ⁺ /K ⁺ -ATPase inhibitor	ered membrane potential/pH → apoptosis in senescent cells		dy. A contest ed correlation al literature (PMID 12856480, 12	RADICTED/contested, identity mismatch flagged); PubMed "di							K ⁺ -ATPase inhibition → membrane depolarization + Na ⁺ /H ⁺ -ex	; cellular senescence/SAIP in glioma is established PD pathol	n, independently reconfirmed: MonarchKG confirms ATP1		man translation data (F3), but the same Na ⁺ /K ⁺ -ATPase/AT	posur e can achieve CNS glioma senescence -cell clearance wh

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	(heart failure, AFib). Also screened Round 2 (cardiac-glycoside			325312) reports elevated "hypotential amplitude digoxin" in PD, but this	goxin Parkinsons on disease" (10 hits, no genuine intervention al study							change r block ad e → acidification, selectively in senescent	ogy theme (independent ly corroborated by the GSK-650394	A3 (HGN C: 801) → "Dystonia 12"/MONDO: 0007496		P1 A3 mechanism is specifically and independently linked	ile staying below the threshold that inhibits neuronal ATP1 A3

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KE EP /R EJ EC T/ H OLD	Reason	Most important remaining inference gap
	eframing): N4 /P 4/ KE EP , M1 /unclear/uncertain, citing the sa			s refers to a disputed, single-group-proposed endo genesis	dy); Monarch digoxin entity (CHEBI: 4551) shows only cardiac							cells in vitro and in vivo (lung fibrosis, tumor xenograft mo	findings above). No CNS/neuronal senescence data in the pri	(= rapid-onset dystonia-parkinsonism, RDP), and literature		to a parkinsonism-causing effect in neurons if CNS exposure	(is of form/dose selectivity window) is entirely untested.

Candidate	Canonical entity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-compound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
	meATP1A3/RDP complication — cited here as SUPPORTED_C			opreno id-pathway metabolite , not administration of the ph	c-disease/intoxication as associations, no PD link							del s) — PMID 31636264 (Nat Commun 2019). Human tra	mary senolytic study.	e (PMID 15260953 et al.) shows reduced neuronal Na ⁺ /		occurs — an unresolved directional safety conflict, compounded	

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-combination functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC/T/HOLD	Reason	Most important remaining inference gap
	CONTEXT only, independently re-verified below for this senol			armaceutical drug — flagged as a name-collision/identity risk,								nslation al: geroprotection/ frailty re purposing trial (P MID 36		K+-ATPase (ATP1A3) function causes a parkinsonism phen		by narrow therapeutic index and uncertain brain penetration.	

Candidate	Candidate	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/RJEC T/HOLD	Reason	Most important remaining inference gap
	ytic framing.			not counted toward N1/P1.								682465).		otype — the identical pump-inhibition direction theapeutically de		Does not meet the Preferred profile (Direction ≠ consistent).	

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
														sir ed for se nol ysi s. Als o, dig oxi n sh ow s po or/ imi ted CS F pe net			

Candidate	Canonical identity	Discovery branch	Target/action	Exact prior PD evidence	Evidence source	N	P	M	F	Direction	CNS	Exact-com pound functional phenotypic	PD relevance of phenotype	Contradictory evidence	KEP/R EJECT/HOLD	Reason	Most important remaining inference gap
														ration at conventional doses (P MID 10 12 99 9).			

Final Counts

- **Total screened: 26** unique candidates (27 raw outputs, Mitoxantrone merged across its two discovery branches)
- **REJECT: 12** — Pyridoxamine, L-carnosine, Dihydromyricetin, Salvianolic acid B, Vitamin K1, PRE-084, Pridopidine, Cutamesine (SA4503), Tubastatin A, Dorsomorphin (Compound C), Dynole 34-2, GSK-650394
- **HOLD: 6** — Bismuth subnitrate/subcitrate, CA77.1, Mitoxantrone (merged), Benzethonium, Pegcetacoplan, Digoxin
- **KEEP: 8** — Oltipraz, Ru265, Neflamapimod (VX-745), Emricasan (IDN-6556), Bis-T-23, ANX005, PMX205, Ruxolitinib

$$12 + 6 + 8 = 26 \checkmark$$

Ranked Shortlist — Strongest Survivors (Preferred profile: P4 AND (M2 or M3) AND (F2 or F3) AND Direction=consistent AND CNS≠fatal_liability AND no direct PD exposure)

Tier 1 — CNS feasibility directly demonstrated, clean direction, own subagent flagged "Preferred":

1. **PMX205** — M2, F2, CNS feasible; strongest exact-compound CNS-penetrant functional dataset of any KEEP candidate (4 independent CNS-disease models); explicitly flagged "best-supported Preferred profile in this branch."
2. **Oltipraz** — M3, F2, CNS feasible; clean N3/P4 novelty, no positive patent signal anywhere; explicitly flagged Preferred.
3. **Ru265** — M3, F2, CNS feasible with directly demonstrated cortical-neuron activity; carries a confirmed off-target excitability/neurotransmission caveat (PMID 38779706) to resolve.
4. **Ruxolitinib** — M2, F2, CNS feasible; FDA-approved (favorable regulatory/PK path); explicitly flagged as meeting the full Preferred profile and the strongest candidate in its branch; chronic immunosuppression risk in elderly PD patients is the main open safety question.
5. **Neflamapimod (VX-745)** — M3, F3 (strong human translational data — Phase 2B trial, plasma p-tau response — but in DLB, not PD); CNS feasible via human CNS trial data.
6. **Emricasan (IDN-6556)** — M2, F2, CNS plausible (rat ischemia model; extensive human liver-safety history); PD-relevance chain assembled from two separate (non-overlapping) literatures — the weakest M-support of this tier.

Tier 2 — Preferred by the literal rule (CNS#fatal_liability includes "uncertain"), but CNS penetration itself is the open question: 7. **ANX005** — M2, F3 (real human CNS-disease trial activity in Huntington's disease); CNS uncertain — large IgG antibody, BBB/CSF penetration unresolved. 8. **Bis-T-23** — M3 (via independent monogenic PD-endocytosis genetics), F2 (real activator phenotype, but in kidney/podocyte disease — cross-tissue extrapolation to synapse is AGENT_INFERENCE); CNS uncertain — zero CNS/BBB data of any kind.

Notable HOLD candidates carrying real signal, not simple rejections:

- **Digoxin** — strong F3 senolytic/human evidence, but Direction=unclear due to an independently reconfirmed ATP1A3/RDP-parkinsonism mechanistic conflict (a genuine safety-relevant complication, not just an evidence gap).
- **Pegcetacoplan** — strong M3/F3 (human clinical data for its own approved indications), but its only validated CNS-adjacent route (intravitreal) has never been shown to cross the BBB systemically — a real class-consistent delivery liability.
- **Mitoxantrone (merged)** — real MCU-PD target rationale, but unresolved exact-compound functional evidence and a well-documented cardiotoxicity liability for chronic dosing.
- **CA77.1** — a genuine evidence gap (identity itself unverifiable through any approved source), not a negative finding.
- **Bismuth subnitrate/subcitrate** — only F1/inferential functional evidence plus a documented human CNS-toxicity liability (encephalopathy at excess exposure).
- **Benzethonium** — F0 screen-hit only, with a confirmed complete absence of any systemic/CNS exposure literature (consistent with its real-world topical-only use).

Senescence/SASP Group — Retry Outcome and Usage

The Senescence/SASP branch (GSK-650394, Ruxolitinib, Digoxin) required three Agent-tool dispatch attempts this session:

1. **Attempt 1** (prior session, before compaction) — failed at the Agent tool's own safety-classifier gate: `claude-sonnet-5[1m] is temporarily unavailable (timed out), so auto mode cannot determine the safety of Agent right now`. No research was performed; no subagent tokens were consumed (the call never reached the subagent).
2. **Attempt 2** (this session, immediately after resuming) — same classifier-timeout error, again before any research began. No subagent tokens consumed.
3. **Attempt 3** (this session, single retry per your instruction not to repeatedly retry) — **succeeded**. Full results are incorporated into the master table above, verbatim from the subagent's output.

Usage for the successful Attempt 3 (the only attempt that actually ran, per the subagent's own self-reported usage block):

- Subagent tokens consumed: **94,880**
- Subagent tool calls made: **60**
- Subagent wall-clock duration: **489,690 ms (~8.2 minutes)**

Attempts 1 and 2 consumed no subagent-side tokens or tool calls — both failed at the orchestration layer before a subagent was ever spawned; their cost was limited to this session's own (parent-level) token usage for constructing and issuing the dispatch, which is not separately itemized by the tool.

No **SCREENING_STATUS = NOT_COMPLETED / TOOL_LAYER_TIMEOUT** fallback was needed — all 26 candidates now carry complete, sourced N/P/M/F/Direction/CNS classifications.

STOP. This is the complete screening table. No Deep Validation has been performed or is planned as part of this task.